

Md Iftekharul Islam

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🌐 iftekharcs.github.io

in LinkedIn

EDUCATION

- University of Tennessee** • Knoxville, TN 2024 – Present
Ph.D. in Computer Science • Ongoing
M.S. in Computer Science • May 2025 • GPA: 4.0/4.0
- Rajshahi University of Engineering & Technology** • Rajshahi, Bangladesh 2014 – 2018
B.S. in Computer Science & Engineering • GPA: 3.76/4.0

PUBLICATIONS

- **I. Islam**, W. Li, X. Wang, S. Li, K. Heaslip. “Heterogeneous Mixed Traffic Control and Coordination.” *IEEE/RSJ IROS*, 2025. [Paper]
- S. Sarker, **I. Islam**, B. Poudel, W. Li. “Beacon: A Naturalistic Driving Dataset During Blackouts for Benchmarking Traffic Reconstruction and Control.” *IEEE/RSJ IROS*, 2025. [Paper]
- C. Raskoti, **I. Islam**, X. Wang, W. Li. “MIAT: Maneuver-Intention-Aware Transformer for Spatio-Temporal Trajectory Prediction.” *IEEE/RSJ IROS*, 2025. [Paper]
- J. Pan, W. Li, W. Liu, **I. Islam**, K. Guo, Y. Yang, S. Zhang, X. Ji, D. Wang. “Mixed Crowd Navigation: Perception, Interaction, Planning, and Control.” *Annual Review of Control, Robotics, and Autonomous Systems*, Vol. 9, 2025. [Paper]

PROJECTS

Heterogeneous Mixed Traffic Control and Coordination (*Research Project*)

- Implemented reinforcement learning framework for coordinating diverse vehicle types at complex intersections with real-world data, achieving 86% reduction in waiting times
- Skills: Python, PyTorch, Reinforcement Learning, SUMO, Multi-Agent Systems, Traffic Simulation

AI Code Detection (*Software Engineering Course Project*)

- Evaluated DetectCodeGPT for distinguishing AI-generated from human-written code across programmer experience levels, achieving AUROC of 0.72 on professional code but 0.47 on student submissions
- Skills: Python, PyTorch, Machine Learning, Statistical Analysis

Subspace Detection for Hyperspectral Image Classification (*Bachelor Thesis*)

- Developed feature extraction method combining Folded-PCA and Cross Cumulative Residual Entropy for hyperspectral image classification with improved accuracy over existing approaches
- Skills: MATLAB, Machine Learning

TEACHING EXPERIENCE

Graduate Teaching Assistant

2023 – Present

- COSC 423/523 - Artificial Intelligence (University of Tennessee – Fall 2025)
- COMP 4030 - Design and Analysis of Algorithms (University of Memphis – Fall 2023)
- Delivered guest lectures for *Reinforcement Learning* course
- Led lab sessions for *Intelligent Transportation Systems* and *Computer Graphics* courses

Lecturer

2019 – 2023

Bangladesh University, Dhaka, Bangladesh

- Taught courses on *Algorithm Design*, *Data Structures*, *Database Systems*, and *Computer Architecture*
- Served as assistant coordinator on the committee developing an outcome-based undergraduate Computer Science curriculum
- Mentored undergraduate student projects

PROFESSIONAL DEVELOPMENT

- **Machine Learning Specialization**, Stanford Online & DeepLearning.AI
- **Certificate in Teaching Methods & Techniques**, Graduate Training Institute, Bangladesh Agricultural University

AWARDS & HONORS

- University Merit Scholarship (2014–2018)